

# Address Translation Vulnerabilities

## ■ The Translation Process

- Intel's Memory Management Unit (MMU) translates virtual addresses to physical addresses through segmentation and paging units.
- This essential mechanism creates attack surfaces that adversaries can exploit to observe memory access patterns.

## ■ Vulnerability Sources

- Page tables, segmentation units, and translation lookaside buffers (TLB) all maintain state information that can leak sensitive data about enclave execution through careful observation and manipulation.

# Page-Fault Driven Attacks

Page-fault driven attacks exploit page fault exceptions to infer sensitive information. Attackers restrict page access by editing page tables, triggering faults when the victim accesses restricted pages. The chronological sequence of page faults reveals memory access patterns.



## Locate Binary

Attacker infers the base address where target application binary is loaded in memory.



## Restrict Access

Attacker modifies page tables to mark specific pages as inaccessible, setting the trap.



## Monitor Faults

Each access to restricted pages triggers page faults, revealing which pages were accessed.



## Extract Secrets

Analyzing fault patterns enables recovery of sensitive data like cryptographic keys.

While limited to page granularity (4KB), these attacks successfully extract large amounts of data including text documents, JPEG outlines, and cryptographic keys from implementations like OpenSSL and Libgcrypt.

# Version Identification Attacks

## Attack Principle

VID attacks identify target code running inside SGX enclaves by exploiting page access patterns, without requiring public access to victim code.

## Information Revealed

Attackers can determine code algorithms, SDK library versions, and configuration information by observing execution patterns across numerous memory access events.

## Security Impact

Experiments demonstrate observable differences in execution patterns between SDK versions and cryptographic algorithm modes, compromising code confidentiality.

# Sneaky Page Monitoring

- SPM attacks achieve finer granularity than traditional page table attacks by combining multiple attack surfaces: page tables, CPU cache, and DRAM. This hybrid approach overcomes limitations of individual attack vectors.
- Key Innovation
  - SPM uses page accessed flags in PTEs to monitor page visits without direct interrupts. Combined with cache-DRAM techniques, it achieves cache-line level observation granularity.
- Efficiency Gains
  - By measuring execution time between entry and exit pages rather than tracking intermediate accesses, SPM reduces interrupts from 71,000 to just 1,300 for EdDSA attacks.